

Abdul Wahid Shah

Ph.D. in Materials Engineering

Current Address Teacher apartment, DGUT, Songhshan Lake, Dongguan, China (513800)

Permanent Address Village Mianawala, Tehsil Isa Khel, District Mianwali, Pakistan (42200)

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E-mail abdulwahid.shah799@gmail.com, **Date of Birth** 1st April 1992 (33 years)

Nationality Pakistani, **Total Publication citations** 170 (google Scholar)



Research Interests

My research interests lie at the intersection of alloy design, advanced manufacturing, materials characterization, and computational tools. I am particularly focused on data-driven alloy design to accelerate the development of next-generation structural and functional materials for applications in automotive, aerospace, and energy sectors. My goal is to integrate machine and deep learning (ML) frameworks with CALPHAD-based thermodynamic modeling, leveraging CALPHAD's predictive capabilities in phase stability and transformation while utilizing ML algorithms to extract meaningful insights from experimental and compositional data.



Skills

Materials Engineering Skills

- Aluminum (Al) and Magnesium (Mg) alloys: Solid knowledge of commercial casting alloys, heat-treatable wrought alloys, and non-heat-treatable wrought alloys.
- Alloy Design for Various Processing Methods: Working experience in designing lightweight alloys optimized for permanent mold casting, high/low-pressure die casting, forging, extrusion, rolling, and metal additive manufacturing (PBF-LB, PBF-EB, and WAAM).
- Heat-treatment: solution-treatment, homogenization, annealing, and aging-treatment.
- Mechanical properties evaluation: tensile, hardness, fracture analysis, low and high cycle fatigue testing.
- Castability and melt quality evaluation: Fluidity test, hot tearing susceptibility, oxidation tendency, and gas porosity (RPT) analysis.
- Characterization techniques: optical and electron microscopy (OM, FESEM, EBSD, and TEM/STEM), X-ray diffraction analysis (XRD), thermal analysis (DSC, TGA), Raman spectroscopy, and X-ray photoelectron spectroscopy (XPS).

- Writing skills: writing for journal publication, Technical Writing, and grant writing.

Computational and Data Analysis Skills

- Technical softwares: Origin, Microsoft Office, ImageJ, FactSage, and JMatPro.
- Programming: Python for data analysis and machine learning applications.
- Machine and deep learning: Experience with machine learning models and deep learning models.
- Data analysis: Skilled in data wrangling and analysis using Pandas, NumPy, and Matplotlib.
- ML libraries: Skills in Scikit-learn, TensorFlow, Keras, and graph libraries like Seaborn for data visualization.
- Technical Expertise: Building and training Machine learning models for prediction tasks and integrating machine learning for alloy design research.
- Basic knowledge of SQL, Apache Spark, and Databricks for big data processing and management.



Work experience

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| 2024-01 | Postdoc Researcher |
| 2025-12 | Department of Materials Science and Engineering, South China University of Technology, Guangzhou, China |
| 2022-08 | Postdoc Researcher |
| 2023.12 | Department of Materials Science and Engineering, Inha University, Incheon, South Korea |
| 2016-02 | Graduate Researcher |
| 2022.08 | Advanced materials and Processing group, Korea Institute of Industrial Technology, Incheon, South Korea. |



Education

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| 2018-02 – | Doctor of Philosophy: Industrial Technology |
| 2022-08 | University of Science and Technology, South Korea - Daejeon |
| | <ul style="list-style-type: none"> • Dissertation: Alloy design, castability, and mechanical properties of Al-Mg-Si alloys with high Mg content |
| 2016-02 - | Master of Engineering: Critical Materials |
| 2018-02 | |

University of Science and Technology, South Korea - Daejeon

- Thesis: *Effect of CaMgSn Ternary Phase on the Aging process and mechanical properties of Mg–Sn–Zn–Ca Alloys*

2009-08 - **Bachelor of Engineering: Materials Engineering**
2014-06 GIK Institute of Engineering Sciences & Technology, Pakistan - Topi



Publications

Published

1. **Abdul Wahid Shah**, Kang Wang, Jabir Ali Siddique, Javaid Hussain, Wenfang Li, Explainable Machine Learning for Predicting Tensile Properties of Aluminum Alloys in the Laser Powder Bed Fusion Process, **Materials Science and Engineering A (2025)**.
2. **Abdul Wahid Shah**, Jabir Ali Siddique, Umar Daraz, Saif Haider Kayani, Javid Hussain, Ghulam Mudassir, Data-driven design of Al-Fe alloys for laser powder bed fusion to enhance recycled aluminum utilization, **Journal of Alloys and Compounds (2025)**.
3. Jabir Ali Siddique, Bong H. Kim, Kwangmin Choi, Naveed Ali, **Abdul Wahid Shah**, Javid Hussain, Ram Song, Hyo Yun Jung, Seong-H. Ha, Young O. Yoon, Hyun K. Lim² and Shae K. Kim, Elimination of stretcher strain marks in high-Mg Al-Mg alloys: deformation behavior and microstructural insights in H16 temper, **Materials Science and Engineering A**.
4. Jian-Xin Sun, Gao-Ming Sun, **Abdul Wahid Shah**, Wen-Fang Li, Kang Wang, Shuo-Xun Jin, Bing-Bing Wan, Da-Xin Zeng, De-Chang Zeng, Wire-powder-arc additive manufacturing of alternating graded TiC and B₄C particles reinforced aluminum matrix composite structures, **Virtual and Physical Prototyping (2025)**.
5. Jabir Ali Siddique, Bong-Hwan Kim, Shahabodin Rafiei, **Abdul Wahid Shah**, Ram Song, Seong-H Ha, Young O Yoon, Hyun K Lim, Shae K Kim, Role of Dislocation Density on the Onset and Intensity of Stretcher Strain Marks in Novel Al-Mg Alloys with High Mg Content, **Journal of Materials Research and Technology (2025)**.
6. Umar Daraz, Hyunkyung Lim, Bong-Hwan Kim, **Abdul Wahid Shah**, Nam-Seok Kim, Shae K. Kim, Silicon addition in novel Almag5 alloy: Balancing castability, strength, and electrical conductivity, **Journal of Alloys and Compounds (2025)**.
7. **Abdul Wahid Shah**, K Wang, JA Siddique, W Li, A comprehensive review of diamond-reinforced metal matrix composites for thermal management in high-performance electronics **Journal of Materials Research and Technology (2024)**.
8. S Jin, Y Li, **Abdul Wahid Shah**, J Sun, B Wan, X Xu, W Li, Insightful investigation for the strengthening mechanisms of Al-Cu alloy prepared by wire arc additive manufacturing, **Journal of Materials Research and Technology (2024)**.
9. Min-Seok Baek, **Abdul Wahid Shah**, Shae-K Kim, Bong-Hwan Kim, Kee-Ahn Lee, High cycle fatigue strength and crack propagation behavior of CaO-AM30 alloy in as-extruded condition, for publication in **Internal journal of metals and materials (2024)**.

10. Min-Seok Baek, **Abdul Wahid Shah**, Young-Kyun Kim, Shae-K Kim, Bong-Hwan Kim, Kee-Ahn Lee, Microstructures, tensile properties, and strengthening mechanisms of novel Al-Mg alloys with high Mg content, **Journal of Alloys and Compounds (2023)**.
11. **Abdul Wahid Shah**, Seong-Ho Ha, Jabir Ali Siddique, Bong-Hwan Kim, Young-Ok Yoon, Hyun-Kyu Lim, Shae K Kim, Microstructure Evolution and Mechanical Properties of Al-Cu-Mg Alloys with Si Addition, **Materials (2023)**.
12. **Abdul Wahid Shah**, Seong-Ho Ha, Jabir Ali Siddique, Bong-Hwan Kim, Young-Ok Yoon, Shae K Kim, Improving the Soundness of Al-Mg-Based Castings Through Chemical Inoculation and Process Optimization, **International Journal of Metal-casting (2023)**.
13. **Abdul Wahid Shah**, Seong-Ho Ha, Jabir Ali Siddique, Bong-Hwan Kim, Young-Ok Yoon, Hyun-Kyu Lim, Shae K Kim, effects of Mg contents on microstructures and mechanical properties of Al-Cu-Mg alloys with improved elongation, **Materials (2023)**.
14. Young-Chul Shin, **Abdul Wahid Shah**, Seong-Ho Ha, Effects of SPD by Biaxial Alternate Forging on the Tensile Properties and Microstructure of AZ31B Magnesium Alloy, **Archives of Metallurgy and Materials (2023)**.
15. **Abdul Wahid Shah**, Seong-Ho Ha, Bong-Hwan Kim, Young-Ok Yoon, Hyun-Kyu Lim, Shae K Kim, *Effect of compositional variation on the microstructural evolution and the castability of Al-Mg-Si ternary alloys*, published in **Metallurgical and Materials Transactions A (2021)**.
16. **Abdul Wahid Shah**, Seong-Ho Ha, Bong-Hwan Kim, Young-Ok Yoon, Hyun-Kyu Lim, Shae K Kim, *Effect of Al₂Ca addition and heat-treatment on the microstructure modification and tensile properties of hypo-eutectic Al-Mg-Si alloys*, published in **Materials (2021)**.
17. **Abdul Wahid Shah**, Seong-Ho Ha, Bong-Hwan Kim, Young-Ok Yoon, Hyun-Kyu Lim, Shae K Kim, *Effect of Si addition on flow behavior in Al-Mg and Al-Mg-Si molten alloys*, published in **Metallurgical and Materials Transactions A (2020)**.
18. **Abdul Wahid Shah**, Seong-Ho Ha, Young-Ok Yoon, Shae K Kim, *Heat treatment of Al-Mg-Si-Mn alloys with varying Mg/Si ratio*, published in **Light metals (2022)**.
19. **Abdul Wahid Shah**, Seong-Ho Ha, Bong-Hwan Kim, Young-Ok Yoon, Hyun-Kyu Lim, Shae K Kim, *Effect of Cu addition on the precipitation hardening and mechanical properties of Al-Mg based cast alloys*, published in **Journal of nanoscience and nanotechnology (2021)**.
20. **Abdul Wahid Shah**, Seong-Ho Ha, Bong-Hwan Kim, Young-Ok Yoon, Hyun-Kyu Lim, Shae K Kim, *Influence of Si content on tensile properties and fractography of Al-Mg-Si ternary alloys*, published in **Journal of nanoscience and nanotechnology (2021)**.
21. **Abdul Wahid Shah**, Hyun-Kyu Lim, Young-Gil Jung, Won-Seok Yang, Seong-Ho Ha, Young-Ok Yoon, Shae K Kim, *Effect of CaMgSn Ternary Phase on the Aging Response of Mg-Sn-Zn-Ca Alloys*, published in **Journal of Korea foundry Society (2018)**.
22. Seong-Ho Ha, **Abdul Wahid Shah**, Bong-Hwan Kim, Young-Ok Yoon, Hyun-Kyu Lim, Shae K Kim, *Oxidation Kinetics of Cu Alloys Containing Alkaline Earth Metals at Low Temperature*, published in **Journal of Nanoscience and Nanotechnology (2021)**.
23. Seong-Ho Ha, **Abdul Wahid Shah**, Bong-Hwan Kim, Young-Ok Yoon, Hyun-Kyu Lim, Shae K Kim, *Effect of Cu addition on oxide growth of Al-7mass%Mg at high temperature*, published in **Archive of Metallurgy and Materials (2021)**.
24. Young-Ok Yoon, Seong-Ho Ha, **Abdul Wahid Shah**, *Influence of Si Addition on oxide Growth of Al-6 mass%Mg Alloy melts*, published in **Archive of Metallurgy and Materials (2021)**.
25. Seong-Ho Ha, **Abdul Wahid Shah**, Bong-Hwan Kim, Young-Ok Yoon, Hyun-Kyu Lim, Shae K Kim, *Effect of Phase Fraction Ratio Between Al₃Mg₂ and Mg₂Si on the Oxidation Behavior of Al-Mg-Si Alloys*, published in **Journal of Nano-electronics and Opto-electronics (2021)**.
26. Seong-Ho Ha, **Abdul Wahid Shah**, Bong-Hwan Kim, Young-Ok Yoon, Hyun-Kyu Lim, Shae K Kim, *Investigation of Oxidation Behaviors of Al-Mg Alloys with a Grain Refiner at High Temperature*,

published in **Journal of Nano-electronics and Opto-electronics (2021)**.

27. Seong-Ho Ha, **Abdul Wahid Shah**, Bong-Hwan Kim, Young-Ok Yoon, Hyun-Kyu Lim, Shae K Kim, Re-Examination of Oxidation in Al-Mg Alloys Containing High Contents of Mg at High Temperatures Using Phase Diagrams, published in **Journal of Nano-electronics and Opto-electronics (2021)**.
28. Seong-Ho Ha, **Abdul Wahid Shah**, Bong-Hwan Kim, Young-Ok Yoon, Hyun-Kyu Lim, Shae K Kim, Effect of Mg on the Oxide Growth of Al-Cu Alloys at Elevated Temperature, published in **Journal of Nano-electronics and Opto-electronics (2021)**

Under-Review

- 1 **Abdul Wahid Shah**, Kang Wang, Yi Huang, Xiaobin Ming, Jabir Ali Siddique, Saif Haider Kayani, Wenfang Li, Enhanced SiC growth on diamond through aluminum-assisted carbon diffusion, **Diamond and related materials**.
- 2 Kang Wang, Wenfang Li, Yi Huang, Xiaobin Ming, **Abdul Wahid Shah**, Zhisheng Zhu, Chao Chen, Yu Guo, Fabrication of Diamond/Aluminum Composites for Electronic Packaging via Electron Beam Powder Bed Fusion, **Additive manufacturing**.
- 3 Javid Hussain, Abdul Wahid Shah, Jabir Ali Siddique, Shoujin Sun, Metallurgical aspects of additively manufactured aluminum alloys: a review, **Journal of Manufacturing process**.



Academic conferences

1. **Inviting speaker** to talk on “microstructures and mechanical properties of high Mg-added Al-Mg alloys” at **KIMM spring 2023** (26-28 April 2023) in Jeju, South Korea.
2. **Oral presentation** on the topic of “Influence of Si content on the castability of Al-Mg-Si based cast alloys”, at **ICAA2020** virtual conference (26–29 October 2020) in Grenoble, France.
3. **Oral presentation** on the topic of “Heat treatment of Al-Mg-Si-Mn alloys with varying Mg/Si ratio”, at **TMS ANNUAL MEETING 2022** (28th February 2022) in Anaheim, USA.
4. **Oral presentation** on the topic of “Effect of CaMgSn Ternary Phase on the Aging process and mechanical properties of Mg-Sn-Zn-Ca Alloys”, at **MEAMT2017** conference (25–27 August 2017) in Busan, South Korea.
5. **Poster presentation** on the topic of “Effect of compositional variation on the precipitation hardening and tensile properties of Al-Mg-Si casting alloys”, at the **ISIMP2021** conference (26–29 November 2021) in Jeju, South Korea.
6. **Poster presentation** on “Elevated temperature stability and deformation behavior of Al-Si-Mg ternary alloys having varies amount of Si and Mg elements”, at the **ICAE2019** conference (5–8 November 2019) in Jeju, South Korea.
7. **Poster presentation** on the topic of “Distribution and precipitation of vanadium (V) in A356 alloy were containing high content of Mg”, at the **ISIMP2017** conference (1–3 November 2017) in Jeju, South Korea.



Research Grant

- The project on “the development of heat-treatable AlMgSiMn based permanent mold (PM) casting alloys (12/2020 ~ 11/2021)” is supported by UST Young Scientist Research Program 2020 through the University of Science and Technology (No. IJ200030).



Achievements

- Awarded Global Talent Visa (F-2-7S) by the Korean government in recognition of research contributions in the field of metallic materials.
- Best student paper award at MEAMT 2017 international conference held in Busan, South Korea.
- Fully funded master and PhD programs at UST Korea, Daejeon, South Korea.
- Fully funded bachelor program at GIK Institute, Topi, Pakistan.